Ranga Rohit Nallamolu

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PROFESSIONAL SUMMARY

Data Engineer with 3+ years of experience building scalable batch and streaming pipelines using Python, PySpark, SQL, Databricks, Kafka, and cloud platforms. Experienced in Delta Lake, CDC, dimensional modeling, workflow orchestration, CI/CD, and automated data quality. Proven ability to optimize data processing, improve pipeline reliability, and deliver financial and operational data for analytics and AI-driven insights.

TECHNICAL SKILLS

Programming Languages: Python, PySpark, SQL, Scala

Cloud & Platforms: Databricks, Azure, AWS, Snowflake

Data Engineering: Apache Spark, Spark Structured Streaming, Kafka, Airflow, dbt

AI/ML: PyTorch, XGBoost, Claude, RAG, Streamlit

BI & Databases: Power BI, SAP Analytics Cloud, PostgreSQL

DevOps: Git, GitHub Actions, Docker, CI/CD

EXPERIENCE

Data Engineer | *BigBlue InfoTech (Client: McKesson), Dallas, TX* **Jan 2025 - Present**

- Engineered PySpark ELT pipelines on Databricks for finance, headcount, and operational data, cutting month-end close prep time 20%.
- Developed Kafka and Spark Structured Streaming pipelines ingesting real-time operational events into Delta Lake with checkpointing.
- Automated data validation and source-to-target reconciliation for nulls, duplicates, schema drift, and freshness, saving 16+ hrs/month.
- Optimized P&L, OPEX, and headcount Spark jobs with partitioning and caching, cutting job runtime and Databricks compute costs 15%.
- Modeled actuals, budget, forecast, and headcount data marts in Delta Lake using MERGE and CDC loads for variance reporting.
- Applied the Claude API on Databricks to tag sentiment and recurring issues in product feedback, loading results into review data marts.
- Orchestrated Airflow and Databricks Workflows with retries and alerting, using GitHub Actions CI/CD and dbt tests for reliable deploys.
- Delivered Power BI dashboards for P&L, OPEX, and headcount to finance leadership, replacing Excel reports and saving 25+ hrs/month.

Graduate Data Engineer | *University of Massachusetts, Lowell, MA* **Dec 2022 - May 2024**

- Ingested student, enrollment, course, and finance data from legacy systems with Azure Data Factory and PySpark on Databricks.
- Structured the data into raw, cleaned, and business-ready Medallion layers on Azure Data Lake, cutting reporting time by 25%.
- Designed star-schema models with SCD Type 2 dimensions, preserving enrollment history for retention and revenue analysis.
- Tuned Spark pipelines using predicate pushdown, Z-ordering, and file compaction, reducing pipeline runtime by 10%.
- Introduced automated data quality checks for completeness, accuracy, freshness, and duplicates, preventing inconsistent data in reports.
- Secured sensitive student data with Databricks Unity Catalog and RBAC, centralizing permissions so each team saw only approved data.
- Created Power BI dashboards for enrollment, course performance, and tuition revenue, presenting insights to department leaders.

Graduate Research Assistant | *University of Massachusetts, Lowell, MA* **Jun 2023 - Nov 2023**

- Consolidated 12 years of metagenomics data from Azure Data Lake into an ML-ready dataset using PySpark on Azure Databricks.
- Built XGBoost, Random Forest, and PyTorch neural network models in Python to predict Type 2 diabetes risk, achieving 85% test accuracy.
- Identified a strong positive correlation (r = 0.70) between *Coprococcus* bacteria abundance and Type 2 diabetes risk.

PROJECTS

AI-Powered Food Delivery Analytics Pipeline *GitHub Link*

- Architected a daily Airflow pipeline loading 10M orders from S3 into Snowflake, with dbt medallion layers and MERGE facts.
- Automated Claude classification of 300K reviews (sentiment, topic, key issue) as a step in the Airflow DAG, feeding a dbt insights mart.
- Built RAG chat and text-to-SQL Streamlit applications to enable natural-language exploration of customer feedback and business metrics.

AI-Driven Stock Trading & Research Platform *Website Link*

- Built a multi-agent AI platform for data-backed stock research, where Claude Haiku/Sonnet Bull/Bear agents debate each trade idea.
- Automated daily research and risk/stop-loss checks on 100+ stocks, cutting evaluation time by 90% and manual monitoring by $\sim$80%.
- Built with Next.js, Node.js, and PostgreSQL on AWS (RDS, Cognito), using live market data from Finnhub and FMP APIs.

EDUCATION

University of Massachusetts Lowell, MA

Master of Science in Computer Science **Sep 2022 - May 2024**

Christ University Bangalore, India

Bachelor of Technology in Computer Science **Aug 2017 - May 2021**